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Agent-based policy analysis: Youth housing affordability and urban market stability

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ABSTRACT

Access to affordable housing for younger generations has become an increasingly urgent challenge in urban policy. However, most evaluations of affordability-focused interventions remain retrospective, lacking forward-looking analyses that anticipate early-stage policy effects under dynamic market conditions. This study employs agent-based modeling (ABM) to assess Taiwan's Youth Housing Loan Program (NYHLP), which aims to reduce financial barriers for young homebuyers while promoting broader market stability. Simulating multiple policy scenarios, the study examines how subsidies, interest-rate changes, and regulatory interventions affect both affordability and systemic dynamics. Findings indicate that demand-side measures enhance access for low- and middle-income buyers but may also trigger inflationary pressures and exclusionary feedback, particularly in supply-constrained contexts. These dynamics, theorized as a feedback-driven affordability trap, highlight ABM's advantage over Dynamic Stochastic General Equilibrium (DSGE) models by capturing endogenous, recursive interactions rather than assuming exogenous equilibrium adjustment. Recent policy adjustments by Taiwan's authorities further corroborate the model's insights, showing how credit liberalization without synchronized supply-side expansion can entrench rather than relieve affordability constraints. The study underscores the importance of adaptive, resilience-oriented housing policy design, offering relevant lessons for other markets facing similar affordability pressures.

1. Introduction

Housing affordability has emerged as a defining global challenge, particularly for younger generations striving to attain homeownership amid rapidly rising property prices and tightening financial conditions. In response, governments across both developed and developing economies have adopted policies to ease financial burdens for young buyers while seeking to correct housing market imbalances. Prominent interventions include housing subsidies, extended loan terms, and interest rate reductions.

Empirical evidence highlights the pivotal role of such initiatives in enhancing social stability and promoting inclusive economic development. In South Korea, for example, financial aid schemes targeting single-person households have demonstrated how subsidies and loan extensions can increase housing access for young adults (Kang & Park, 2024). Singapore's public housing programs combine generous subsidies with favorable loan terms, successfully enabling young families to transition into homeownership (Galster & Lee, 2020). In the United States, a mix of tax incentives and reduced mortgage interest rates has eased the burdens on first-time homebuyers, thereby encouraging broader participation in the housing market (Deng et al., 2024).

Against this international backdrop, Taiwan's New Youth Home Loan Program (NYHLP), launched in 2023, presents a timely and instructive case. The NYHLP is a government-backed mortgage subsidy targeting first-time homebuyers under the age of 35, designed to improve housing affordability among younger generations. The program offers increased loan limits, extended repayment schedules, and below-market interest rates to young prospective homeowners. In doing so, it not only aims to expand youth access to homeownership but also to stimulate the broader housing market, a policy strategy similar to earlier interventions in mainland China. However, the NYHLP remains in its initial state, and its long-term implications are not yet fully understood. This context provides a critical opportunity to assess its potential effects through predictive, dynamic analysis.

While most of the existing research on youth housing policy relies on post-implementation evaluations using historical data, relatively few studies investigate policy effects at the early-implementation stage. Moreover, traditional econometric models often struggle to represent

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the dynamic and adaptive behavior of market participants reacting to evolving policy conditions. Addressing this gap, this study applies an agent-based modeling (ABM) framework to simulate the likely effects of Taiwan's NYHLP under various policy scenarios.

ABM allows the exploration of interactions among key stakeholders—including homebuyers, developers, financial institutions, and government agencies—within a simulated housing market environment. By capturing emerging behavior and feedback effects, the approach generates insights into how youth-targeted policies shape affordability and market stability over time.

This study adopts a forward-looking orientation to anticipate both market dynamics and potential policy risks as the NYHLP unfolds. In doing so, it seeks to equip policymakers with actionable foresight for identifying unintended consequences and fine-tuning policy instruments in real time.

The study's central objective is to evaluate the NYHLP's likely outcomes across two primary dimensions: (1) easing financial pressure on homebuyers, and (2) stimulating broader housing market activity. Through ABM simulations, the study offers evidence-based recommendations to strengthen program design and implementation. Its emphasis on scenario-based analysis is particularly relevant for emerging economies, where housing policy must often operate under conditions of uncertainty and limited historical data.

Beyond framing youth housing affordability as an economic challenge, this study highlights its broader implications for urban governance and sustainable development.

It contributes to the housing policy literature in three principal ways. First, it advances theoretical understanding by modeling the unintended consequences of demand-side subsidies in supply-constrained urban markets, particularly the feedback loops between affordability policies and speculative price inflation. Second, it provides empirical foresight by simulating the early-stage impacts of the NYHLP using ABM, an underutilized method in youth housing policy research. Third, it introduces a flexible ABM framework that incorporates adaptive credit constraints and policy trigger mechanisms, offering a practical tool for anticipating risks in a data-scarce or fast-evolving housing market.

The remainder of this study is structured as follows: Section 2 reviews the literature, identifying key theoretical frameworks and research gaps. Section 3 outlines the ABM methodology, including model design and implementation. Section 4 presents the simulation results. Section 5 discusses the findings in relation to existing evidence, while Section 6 offers policy implications and suggests directions for future research.

2. Literature review

2.1. Housing loan policies and homebuyer behavior

Housing loan policies play a significant role in shaping homebuyer behavior by enhancing purchasing power and lowering entry barriers to the housing market. Empirical studies indicate that instruments such as reduced interest rates, higher loan ceilings, and extended repayment periods can increase homeownership rates, particularly among middle- and high-income groups (Wang, Chen, & Tsai, 2022a, 2022b). In Taiwan, such measures have been central to addressing affordability concerns amid soaring property prices. However, lower-income households continue to face persistent obstacles, including stringent credit evaluations and high down payment requirements, which limit their ability to benefit from these policies (Huang & Wang, 2021).

Similar challenges have emerged in other high-demand housing markets. In South Korea and Singapore, for instance, affordability-focused interventions, such as housing subsidies and favorable loan terms, have yielded mixed results. Research by Lee and Kim (2020a, 2020b) shows that, in South Korea, subsidies increased ownership rates but also contributed to price inflation in urban centers. Likewise, Singapore's public housing policies have improved affordability but required tight regulatory oversight to mitigate speculative activity (Chen & Tan, 2021).

Comparison with Mainland China further highlights the shared policy challenges in targeting young and first-time homebuyers. Studies show that despite intentions to assist lower-income groups, housing policies, including reduced down payments and longer loan terms, disproportionately favored wealthier buyers, thereby intensifying demand-driven price escalation in the urban market (Zhang & Li, 2019a, 2019b). These findings underscore the importance of combining demand-side interventions with adequate supply-side responses to ensure both affordability and long-term market stability.

The COVID-19 pandemic has added further complexity to global housing policy. In response, countries such as the United States, South Korea, and Taiwan introduced housing loan subsidies as part of economic recovery strategies. However, emerging evidence suggests that in the absence of corresponding supply-side adjustments, these interventions may exacerbate price pressures in already competitive markets (Bangura, Lee, & Schafer, 2023a, 2023b). For instance, Alharbi (2024a, 2024b) reports that while United States subsidies have improved affordability for targeted groups, they have also contributed to inflationary trends in high-demand regions. These global experiences reinforce the relevance of Taiwan's NYHLP as a case study with broader implications for both emerging and advanced economies.

2.2. Applications of ABM in economic and housing market research

ABM has emerged as a powerful methodological tool for analyzing housing market dynamics and evaluating policy interventions. Unlike conventional econometric models, ABM allows for the simulation of complex, non-linear interactions among heterogeneous agents, such as homebuyers, developers, banks, and government institutions. This capability makes ABM particularly valuable for uncovering the unintended consequences of policy shifts and for anticipating emergent behaviors across market segments.

In Taiwan, Chen et al. (2023) demonstrated the potential of ABM to assess the effects of interest rate subsidies and extended loan terms on affordability and market stability. Similarly, Lee and Kim (2020a, 2020b) employed ABM to simulate housing policy outcomes in South Korea, finding that while subsidies improved ownership rates, they also encouraged speculative investment in high-demand areas. Parallel findings have been reported in Mainland China, where Gao, Fan, and Pang (2025) used ABM to evaluate the impact of credit control. Their results showed that poorly calibrated policies risk excluding lower-income buyers while simultaneously heightening price volatility.

Recent advancements in ABM have integrated behavioral economics principles, enabling researchers to analyze how cognitive biases affect agent behavior. For example, Su and Qian (2023) examined the influence of over-optimism and loss aversion on buyers' responses to subsidy programs, offering insights into behavioral dynamics often missed in conventional models. In Taiwan, ABM simulations by Chen (2020a, 2020b) and Wang (2022) projected that while lower interest rates and government subsidies improved affordability, they also risk accelerating property price inflation in urban centers in the absence of concurrent supply-side expansions.

2.3. Homebuyer decision-making theories

Homebuyer decisions are shaped by both economic conditions and psychological factors, particularly in markets characterized by rising prices and restricted credit access. Behavioral economics provides a valuable framework for understanding how cognitive biases—such as over-optimism, risk aversion, and loss aversion—influence home purchasing behavior. For instance, Lee, Park, and Kim (2020) found that risk aversion becomes more pronounced under rising interest rates and tighter credit regulations, prompting lower-income households to delay purchases.

Price expectations and macroeconomic conditions also significantly influence buyer behavior. Liao and Zhou (2021) demonstrate that during interest rate hikes, lower-income buyers tend to withhold purchasing decisions, even in the presence of subsidies. Conversely, favorable loan terms and subsidy announcements often prompt first-time buyers in Taiwan to accelerate purchase decisions due to anticipated future price increases (Chen et al., 2024). This sense of urgency can amplify short-term demand and contribute to speculative momentum, highlighting the need for policy mechanisms that stabilize buyer expectations and promote affordability.

2.4. Research focus and contributions

This study contributes to the urban housing literature by examining the effects of Taiwan's NYHLP on both homebuyer behavior and housing market dynamics, with an emphasis on differentiated outcomes across income groups. By employing an ABM framework, this study offers a dynamic assessment of key policy scenarios, including changes in interest rate, loan terms, and credit access restrictions.

The findings offer three principal contributions:

1. **Theoretical contribution:** It extends the application of ABM to model the simulate the complex interplay between housing policies, instruments, and market dynamics, particularly in volatile urban housing environments.
2. **Empirical contribution:** It provides evidence of the dual impacts of demand-side measures—namely, improving affordability while inadvertently driving price inflation—and highlights the important role of integrating supply-side responses.
3. **Policy guidance:** It proposes practical strategies for refining loan subsidies and credit policies to enhance inclusivity for lower-income buyers while mitigating speculative risks.

By linking housing affordability initiatives to broader urban governance challenges, this study contributes to a deeper understanding of how adaptive, evidence-based policies can promote inclusivity, sustainability, and equitable urban development.

3. Methodology and data sources

This study adopts an ABM approach to simulate the interactions among key stakeholders in Taiwan's housing market, with particular focus on the Youth Housing Loan Program (NYHLP) initiated in 2024. Rooted in complexity science and behavioral economics, ABM enables the modeling of heterogeneous agents and non-linear interactions, offering granular insights into how individual decisions—such as mortgage applications, load approvals, and supply adjustments—collectively shape affordability and market stability.

In contrast to traditional econometric models, ABM captures emergent phenomena and recursive feedback effects across agents, including homebuyers, banks, developers, and regulatory authorities. This makes ABM especially suited for examining how staged policy interventions—such as interest rate caps or subsidy thresholds—affect systemic resilience under dynamic conditions. Through multiple simulated scenarios, the model reveals intended policy outcomes and potential side effects, such as speculative inflation or delayed supply response.

The research methodology consists of four main components: model design, data acquisition, scenario construction, and simulation execution. These components are detailed in the following sections to ensure methodological transparency and policy relevance.

3.1. Research model design

The ABM is designed to replicate the decision-making processes and interactions of four primary agents—homebuyers, banks, developers, and government authorities—within the institutional context of the NYHLP. Each agent class is defined by behavioral rules and attributes calibrated to empirical data, ensuring a realistic response to both market and policy changes.

- **Homebuyers** are characterized by income, savings, creditworthiness, and down payment capacity. Their purchase decisions are modeled as a function of affordability (price-to-income ratio), expected price growth, and access to financial incentives.
- **Banks** apply lending criteria based on interest rates, borrower risk profiles, and credit thresholds. The loan approval probability for buyer i is calculated via a logistic function:

$$L_{\mathrm{p}}(i) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 I_i - \beta_2 R_t)}}$$

where I_i denotes buyer income and R_t is the prevailing interest rate. The parameters β are estimated from historical loan approval data (2022−2023).

- **Developers** adjust housing supply and pricing in response to market signals. Prices evolve dynamically as a function of excess demand:

$$P(\mathrm{t}) = P(t-1) \times \left[1 + \alpha\left(\frac{D(t)}{S(t)} - 1\right)\right]$$

where $D(t)$ and $S(t)$ represent current demand and housing stock, respectively, and α is a price elasticity coefficient derived from 2018 to 2022 transaction records.

- **Government authorities** implement interventions (e.g., interest rate caps, down payment subsidies) that influence borrower affordability thresholds and adjust bank lending behavior.

A distinguishing innovation of this model lies in its endogenous feedback loop: subsidies lower borrowing costs, which increase loan approval rates, drive up demand $D(t)$, and subsequently prompts developers to raise prices $P(t)$. This recursive cycle may partially offset the intended benefits of affordability improvements by inducing speculative responses.

3.1.1. Model innovations

In contrast to earlier ABMs that treat policy parameters as static inputs, this model incorporates temporal policy triggers aligned with the phased rollout of the NYHLP (2023–2026). Credit constraints and loan approval conditions evolve in response to macroeconomic changes and policy updates. Moreover, supply constraints are modeled as responsive developer behavior, rather than fixed limits, allowing the simulation to capture speculative inflation patterns when demand growth outpaces supply.

Fig. 1 visualizes these dynamic interactions, illustrating how policy levers (e.g., interest rate caps or subsidies) influence banks' credit allocation, homebuyer affordability, and developer pricing strategies. These interdependencies collectively shape broader outcomes such as price as loan approval rates, price volatility, and shifts in affordability.

Compared to conventional ABMs that assume static credit rules and fixed pricing behaviors (e.g., Lee & Kim, 2020a, 2020b; Su & Qian, 2023), this model incorporates dynamic responses across agents and policy phases. It captures emergent phenomena such as affordability traps and speculative bubbles under phased government interventions. This comprehensive framework enables simulation of both short-term cumulative long-term effects, offering actionable insights for achieving balanced and sustainable housing conditions.

This system diagram illustrates the interdependence among government, banks, homebuyers, and developers within the NYHLP framework. It highlights how policy instruments, such as interest rate caps and down payment subsidies, affect buyer behavior and elicit recursive responses from banks and developers. The diagram also maps

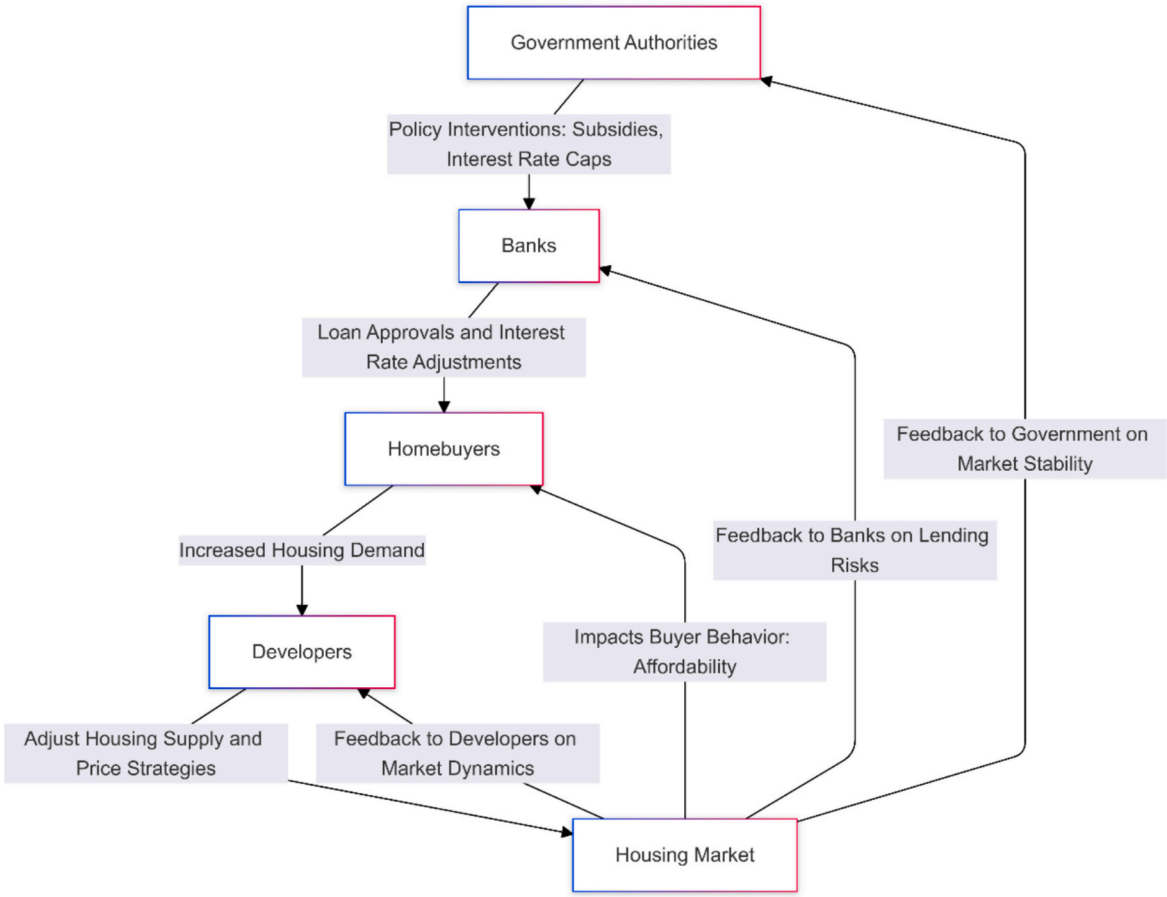

Fig. 1. Dynamic interactions and feedback mechanisms in Taiwan's housing market under the NYHLP framework.

macro-feedback to government authorities, underscoring the importance of endogeneity, systemic volatility, and adaptive policy design.

3.2. Data sources

To ensure the validity and contextual accuracy of the model, this study integrates a multidimensional dataset comprising economic, demographic, financial, and policy-related variables. Data sources were selected based on three key criteria: (1) institutional credibility (e.g., central banks, national statistical bureaus, and official financial bulletins); (2) temporal alignment with the implementation window of Taiwan's Youth Housing Loan Program (NYHLP) from 2022 to 2024; and (3) relevance to agent-level decision-making in the housing market (Table 1).

To validate data accuracy, the study conducted cross-referencing across official and industry sources, including the Taiwan Real Estate Transaction Database and quarterly construction bulletins. For Instance, average housing prices and loan approval rates in 2023Q1 by the Transaction Database deviated by less than 3 % from Central Bank figures, demonstrating internal consistency. Additionally, historical elasticity estimates (2018–2022) were used to calibrate developer behavior, enhancing the reliability of model assumptions.

Table 1
Data sources and key attributes for ABM framework development.

Data category	Key attributes	Source
Economic indicators	Housing prices, interest rates, loan terms	Taiwan central bank and financial reports
Demographic information	Income distributions, household sizes, ownership rates	National statistics and surveys
Banking data	Loan approval rates, credit policies, risk frameworks	Financial institution disclosures
Developer data	Housing supply, construction timelines, market absorption rates	Industry reports and historical data
Policy variables	Loan subsidies, interest rate caps, credit restrictions	NYHLP amendments and policy documents

To improve representativeness and generalizability, this study introduces a cross-regional benchmark analysis. Comparative benchmarks were drawn from Seoul, Singapore, and Shanghai, three high-density urban markets sharing similar affordability pressures and active government interventions. For example, Taiwan's price-to-income ratio (PIR) averaged 15.2 in 2023, closely mirroring Seoul's 16.4 and exceeding Singapore's median PIR of 13.5 (MLIT Korea, 2023; URA Singapore, 2023). Mortgage interest rates in Taiwan (1.7 %) also fall within the 1.5 %–1.9 % range observed in Korea and Singapore, while first-time loan approval rates (45 %–50 %) and subsidies proportions (10 %–15 %) are similarly aligned.

Although real-time microdata (e.g., transaction-level buyer behavior) is unavailable, the model is built on nationally representative statistics, validated assumptions, and sensitivity analyses (see Section 4.5). Parameter estimates were triangulated across multiple datasets, and simulation calibration was employed to mitigate selection bias and enhance robustness under conditions. This design measures improve both internal validity and cross-market transferability, reinforcing the model's relevance for other urban systems facing youth affordability challenges.

3.3. Scenario settings

The study examines multiple policy scenarios to isolate and assess

the effects of individual interventions on housing affordability and market behavior. Each scenario simulates a unique combination of government policy tools and macroeconomic conditions, offering insight into both direct impacts and emergent secondary effects.

Key scenarios include:

- Stricter Lending Regulations: Tighter loan approval criteria for multi-property owners to discourage speculative investment.
- Enhanced First-Time Buyers Subsidies: Increased government support for initial down payments to expand affordability for low-income households.
- Dynamic Interest Rate Adjustments: Periodic caps or reductions in interest rates to improve borrowing access while managing excess demand.

For example, one scenario models the effects of expanding first-time buyers subsidies, which increase loan approvals and intensify housing demand. Another scenario explores credit control aimed at limiting speculative purchases. While such controls may stabilize prices, they could also inadvertently reduce accessibility for financially constrained buyers. These scenarios are designed to offer actionable trade-off analyses for policymakers seeking to balance equity and market stability.

3.4. Simulation procedure

The simulation procedure begins with model calibration using baseline conditions from 2022 to 2023, incorporating interest rates, loan thresholds, housing inventory levels, and developer price elasticity. The simulation spans a four-year horizon (2023–2026), aligning with the phased implementation of the NYHLP. Each run captures both short-term responses and long-term market adaptations.

3.4.1. Policy trigger mechanism

Unlike conventional ABMs that rely on static input, this model adopts a conditional trigger-based intervention structure. Policy levers—such as interest rate caps or subsidy adjustments—are activated based on real-time market indicators. For instance, when simulated housing price growth exceeds 8 % for two consecutive periods, a subsidy reduction is triggered, thereby modifying the affordability threshold and altering buyer-bank interactions.

3.4.2. Credit constraint dynamics

Loan approval for a homebuyer agent i is determined by a logistic regression function:

$$P_{approval.i} = \frac{1}{1 + \exp[-(\beta_0 + \beta_1 \cdot I_i - \beta_2 \cdot R_t)]}$$

where I_i represents the buyer's income and R_t is the prevailing mortgage interest rate. Coefficients β are estimated using empirical lending data from 2022 to 2023, reflecting institutional credit polices under fluctuating macroeconomic conditions.

3.4.3. Endogenous developer response

Developer pricing strategies are modeled using a demand–supply elasticity function:

$$P_t + 1 = P_t \cdot [1 + \alpha \cdot (D_t - S_t)/S_t]$$

Here, D_t and S_t denote demand and supply at time t, and α is the price elasticity parameter estimated from pre-policy market data (2022–2023). This mechanism allows the model to stimulate speculative inflation triggered by policy-induced demand shocks in supply-constrained environments.

3.4.4. Simulation metrics and output

Each simulation tracks a set of key metrics, including: loan approval rates, home purchase volumes, price volatility and trend shifts, and affordability ratios by income group. Agents iteratively adjust their strategies based on market signals and policy conditions. Post-simulation analysis aggregates results to evaluate how different scenarios shape housing access and market resilience.

3.4.5. Robustness testing

To ensure the reliability of outcomes, the model employs Monte Carlo simulations with 100 iterations per scenario, along with sensitivity tests on critical parameters such as size, price elasticity, and credit thresholds. These robustness checks generate confidence intervals and average trajectories for key outputs, enhancing analytical rigor and external validity (Table 2).

3.4.6. Methodological contribution

Compared to prior ABMs (e.g., Lee & Kim, 2020a, 2020b; Su & Qian, 2023) that rely on fixed agent rules, this simulation introduces three key innovations: dynamic credit thresholds, endogenous pricing feedback, and adaptive policy interventions. These mechanisms enable the model to capture emergent risks—such as affordability traps and speculative booms—while maintaining policy realism. The framework is tailored for early-stage scenario analysis in data-scarce, high-volatility environments typical of urban housing policy innovation.

To aid reader comprehension, Box 1 summarizes the model's key behavioral mechanisms.

3.5. Model implementation and calibration procedures

While Section 3.1 presented the conceptual structure of agents and policy mechanisms, this section provides the technical implementation and calibration details necessary for transparency and replicability.

The simulation is implemented in NetLogo 6.3, a widely adopted platform for modeling complex adaptive systems with heterogeneous agents. The model simulates interactions among four agent classes—homebuyers, banks, developers, and government authorities—across the 2023–2026 period, with each time step representing a quarterly decision cycle. To reflect real-world uncertainty, the model integrates both deterministic rules and stochastic variability (e.g., income shocks, loan rejection.)

3.5.1. Simulation execution logic

Each simulation run follows a four-step sequential process:

1. **Market initialization:** The model loads baseline—interest rates, property prices, credit ratios—using official 2022–2023 housing data.
2. **Policy evaluation trigger:** Conditional policies (e.g., subsidy reduction or interest caps) are activated when market indicators surpass predefined thresholds. For example, if housing price inflation exceeds 8 % over two quarters, subsidy reductions are triggered.
3. **Agent actions:**
   - **Homebuyers evaluate affordability and decide whether to enter the market.** A purchase attempt is initiated only if the monthly mortgage burden is under 35 % of disposable income and the expected resale gain (based on a two-quarter moving average)

Table 2
Simulation outcomes under different policy scenarios.

Policy scenario	Loan approval rate	Housing demand	Price trend	Developer response
High subsidies, low rates	Increased	Strong	+5 % to +10 %	Supply expansion
Reduced subsidies, high rates	Decreased	Weak	−5 % to −7 %	Supply contraction
Balanced policies	Stable	Moderate	Stable	Steady supply

Box 1
Mathematical representation of key agent behaviors and feedback mechanisms.

1 Loan approval – homebuyers:
P_approve = 1 / (1 + exp.⁻(β₀ + β₁·I_i + β₂·R_t)).
(Logistic function of income and interest rate).
2 Developer Pricing Rule:
$P_{t+1} = P_t \cdot [1 + \alpha \cdot (D_t - S_t)/S_t]$.
(Endogenous response to demand-supply gap).
3 Policy Trigger:
If price ↑ ≥ 8 % for 2 periods → subsidy cut.
(Replicates Taiwan's NYHLP phase-out logic).

Refer to Sections 3.1–3.4 for detailed parameter explanations.

exceeds 3 %. These heuristics reflect Taiwan's culturally conservative mortgage behavior.

- Banks evaluate applicants using the logistic function:

$$P(approval_i) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 I_i + \beta_2 R_t)}}$$

where I_i denotes household income and R_t the prevailing interest rate. Parameters β are derived from Taiwan's Financial Supervisory Commission (FSC) and Central Bank mortgage data (2022–2023).

- Developers update housing prices based on demand–supply elasticity:

$$\Delta P_t = \alpha \cdot \left(\frac{D_t - S_t}{S_t}\right)$$

where D_t and S_t represent current demand and supply, and α is calibrated using transaction-level data from the Ministry of the Interior.

4. **Market update and data logging:** All outcomes, including price adjustments, purchase completions, and loan approvals, are recorded per time step. The model uses a rotating synchronous update mechanism, ensuring that decisions by one agent class (e.g., developers) immediately inform others within the same simulation cycle.

3.5.2. Parameter calibration

Key parameters were derived from empirical sources (Table 3):

- β_1 and β_2 were estimated via logistic regression, repressing loan approval outcomes against income brackets and lending rates using the FSC-certified mortgage sample (2022–2023).
- α (price elasticity) was estimated through ordinary least squares regression of quarterly price movements against demand-supply gaps using Ministry transaction data (2018–2022).

Table 3
Parameter calibration and data sources for simulation scenarios.

Parameter	Description	Data source
β_1, β_2	Credit approval sensitivity	Taiwan FSC mortgage database (2022–2023)
α	Developer price elasticity	Transaction-level price dynamics, MoI real estate registry
Threshold: Price +8 % for 2 periods	Trigger condition for subsidy reduction	Central Bank Policy Reports, 2023
Initial interest rates	Baseline lending conditions	Central Bank monthly bulletins, 2022–2023

3.5.3. Stochastic settings and robustness

To account for behavioral variability and ensure robustness, the model performs 100 Monte Carlo iterations per scenario. Each run randomly sampled household characteristics from Taiwan's demographic distributions (sourced from the National Development Council census). Aggregated outputs are reported with 95 % confidence intervals, providing a range for evaluating policy effectiveness under uncertainty.

3.5.4. Code availability

A cleaned NetLogo script and full codebook are available upon request to support replication or adaptation to other East Asian housing systems.

3.6. Policy scenarios and design rationale

To evaluate how housing affordability and market stability respond to evolving financial and regulatory conditions, three distinct policy scenarios were simulated: **(1) Baseline Continuation, (2) Gradual Tightening,** and **(3) Shock Tightening.** These scenarios are empirically grounded, drawing from observed and projected developments in Taiwan's housing and financial policy between 2023 and 2025.

Scenario descriptions

- **Baseline** Continuation reflects NYHLP's 2023 launch phase, characterized by a 2.0 % preferential interest rate and full subsidies for first-time buyers.
- **Gradual** Tightening models 2024 conditions, incorporating increased scrutiny on borrowers, informal lending quota constraints, and incremental subsidy tapering in response to macroprudential shifts.
- **Shock Tightening simulates projected** 2025–2026 conditions, including the scheduled expiration of NYHLP subsidies (by July 2026) and reductions in NYHLP lending targets by state-owned banks (see China Times, July 21, 2025).

Each scenario operationalizes a distinct configuration of policy

Table 4
Summary of scenario logic.

Scenario	Subsidy policy	Credit condition	Expected market effect
Baseline	High subsidy	loose credit	Demand surge, mild inflation
Gradual tightening	Moderate subsidy	Stricter approval criteria	Dampened demand, stabilized prices
Shock tightening	Subsidy phaseout	Constrained lending	Risk of affordability trap, demand drop

instruments and institutional responses in Tables 4 and 5. This enables analysis of how subsidy-induced demand interacts with evolving credit frictions, potentially amplifying or mitigating feedback effects.

By formally aligning simulation design with concrete policy markers, this section addresses concerns regarding scenario arbitrariness and establishes a stronger theoretical-policy nexus for interpreting downstream results.

4. Results analysis

This study utilizes an advanced ABM framework to simulate interactions among key stakeholders in Taiwan's housing market, with a particular emphasis on the 2024 NYHLP. ABM. Grounded in complexity science and behavioral economics, ABM enables the representation of heterogeneous agents and non-linear decision-making processes. Unlike conventional econometric approaches, ABM allows for the dynamic modeling of emergent behavior across interconnected actors, making it particularly effective in evaluating the systematic effects of housing policy interventions.

This section presents the simulation results across multiple policy configurations, interpreting their implications for youth affordability, urban housing access, and broader market stability. The analysis is organized progressively: first providing an overview of general trends, followed by a breakdown of interest rate effects, temporal dynamics, and scenario-based comparisons.

By integrating perspectives from systems theory, these simulations offer insight into how policies, market signals, and agent-level decisions interact to produce macro-level outcomes. The findings reveal both expected and unintended consequences—such as price inflation and affordability traps—under varying regulatory and financial environments.

4.1. Simulation results overview

The simulation reveals distinct market responses to changes in subsidies, interest rates, and credit regulations. In a scenario characterized by generous subsidies and low interest rates, lower-income and first-time buyers show markedly higher participation rates. These favorable financing conditions significantly improve affordability, especially for younger cohorts facing credit constraints.

However, the increase in demand generates upward pressure on prices, especially in supply-constrained urban areas. Developers respond to the surge by raising prices, creating a feedback loop in which improved affordability leads to greater demand, thereby inflating prices and partially eroding the policy's intended benefits.

Conversely, scenarios with reduced subsidies and higher interest rates dampen both demand and price growth. Buyer participation drops, especially among lower-income households, while overall price trends stabilize or decline.

These patterns underscore the importance of calibrated interventions that consider both the demand-side benefits and supply-side consequences. Table 6 summarizes homebuyer behavior and price dynamics under three key policy scenarios.

Table 6
Homebuyer behavior across policy scenarios.

Policy scenario	Total homebuyers	Loans approved of homebuyers (%)	Average home price (million NTD)	Average interest rate (%)
Scenario 1: high subsidy, low rates	s	45	16.5	1.775
Scenario 2: no subsidy, rising rates	200	30	15.8	1.865
Scenario 3: high subsidy, strict lending	200	35	16.2	1.825

4.2. Effects of interest rate adjustments on loan approvals

Interest rate emerges as a central determinant of loan accessibility. Lower rates lead to significantly higher approval probabilities, particularly for younger buyers and those in lower income brackets. When combined with subsidies, these conditions facilitate broad market participation, reinforcing the affordability goals of the NYHLP.

In contrast, interest rate hikes result in sharp reductions in loan approvals, disproportionately excluding financially vulnerable groups. Even a modest increase in borrowing costs reduces eligibility rates, underscoring the fragility of access for those at the margins of the credit system.

Fig. 2 illustrates how loan approval evolves under three policy scenarios over a 36-month horizon.

This line graph compares approval trajectories under Scenario 1 (High Subsidy, Low Rates), Scenario 2 (No Subsidy, Rising Rates), and Scenario 3 (High Subsidy, Strict Lending). While subsidies improve access in Scenario 1 (orange), restrictive credit filters in Scenario 3 (yellow) sharply suppress approvals. The visualization demonstrates that the effectiveness of subsidies is contingent on broader credit policy design.

4.3. Simulation timeframe and price trends

Over the 36-month projection period (2023–2026), the model tracks evolving approval rates and price movements across Taiwan's major metropolitan area. The results show that sustained subsidies and favorable interest rates continue to elevate loan approvals across income groups. However, these benefits are spatially unseen: urban centers with tight supply conditions, such as Taipei, Taichung, and Kaohsiung, experience disproportionately sharp price increases.

Fig. 3 presents average house price trends by region under the different policy configurations.

The graph tracks simulated housing prices across six major cities, normalized to a baseline of 100. Taipei and Taichung show the steepest increases under sustained subsidy regimes, while smaller urban markets remain relatively stable. These results highlight regional divergence in policy effects and suggest the need for geographically targeted housing interventions.

Table 5
Simulation scenario design and policy anchors.

Scenario	Interest cap	Subsidy timeline	Credit constraint level	Policy anchor source
Baseline	2.0 %	Full (2023–2026)	Low (Lenient approvals)	NYHLP Launch Circular (MOF, 2023)
Gradual tightening	2.2 %	Partial by 2025	Moderate (Bank-level filters)	Central Bank Memo on Mortgage Risk (2024)
Shock tightening	2.5 %+	Ends July 2026	High (Quota reductions)	China Times, July 21, 2025; CBC Reports (2025)

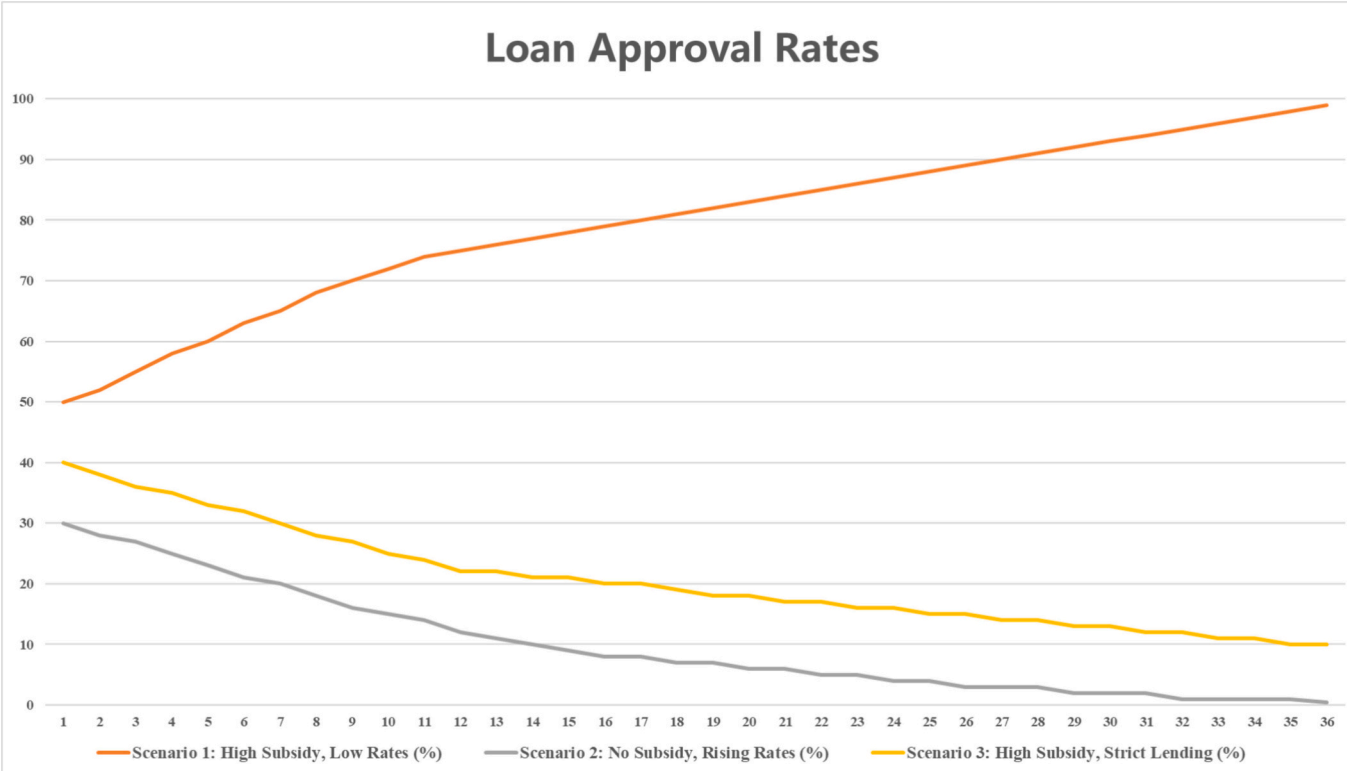

Fig. 2. Loan approval rates over 36 months across three policy scenarios.

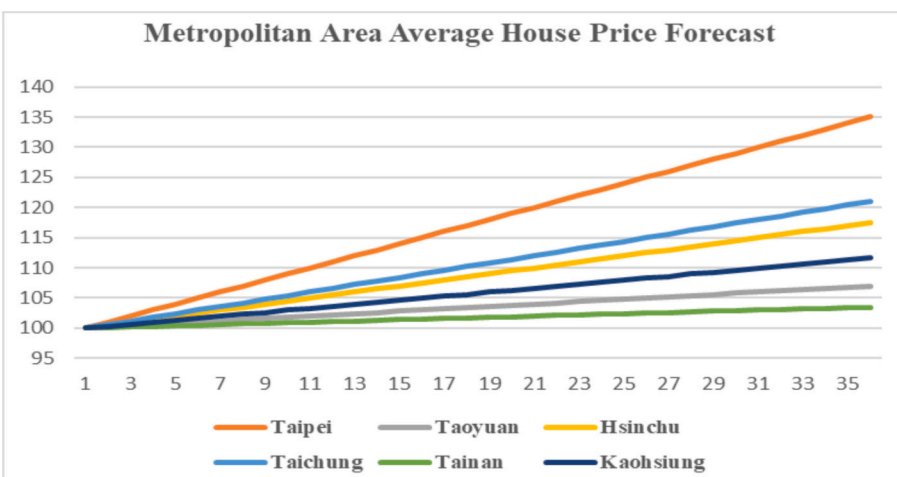

Fig. 3. Average house price forecast in Taiwan's major metropolitan areas (2023–2026).

4.4. Comparative scenarios analysis: demand-side versus supply-side tensions

Three policy scenarios were evaluated to assess how different combinations of subsidies and interest rate setting affect buyer and developer behavior:

- **Scenario 1 (High Subsidy, Low Rates):** Produces the largest increase in loan approvals and participation, especially among lower-income buyers. However, rapid demand growth triggers notable price inflation as developers capitalize on increased purchasing power.
- **Scenario 2 (Moderate Subsidy Rates):** Strikes more balanced outcomes. Access improved without triggering substantial price acceleration, indicating greater long-term sustainability.
- **Scenario 3 (Low Subsidy, High Rates):** Leads to reduced demand and slower price growth. While this curbs inflation, it significantly limits access for marginalized groups, highlighting the exclusionary effects of credit tightening.

Table 7 summarizes the comparative effects of each policy scenario on buyer activity, developer pricing behavior, and market equilibrium.

These results reveal the trade-offs inherent in housing policy design. Demand-side tools can temporarily improve affordability but may also induce inflationary dynamics if not matched by corresponding supply-

Table 7
Impact of government policies on buyer and developer behavior across policy scenarios.

Policy measures	Homebuyer behavior	Developer response	Market supply-demand balance
High subsidy, low interest rates	Increased purchasing intent, higher loan approval rates	Increased housing supply, higher prices	Strong demand, growing supply
No subsidy, rising interest rates	Decreased purchasing intent, fewer loan approvals	Reduced supply, lower prices	Weakened demand, reduced supply
Low subsidy, high interest rates	Sharp decline in loan approvals, reduced demand	Limited response, slower price growth	Weak demand, supply stagnation in low-demand areas

side responses. Conversely, restrictive financial settings reduce speculative pressure but risk reinforcing structural exclusion.

4.5. Sensitivity of simulation assumptions

To evaluate the robustness of the simulation outcomes under parameter uncertainty, a series of sensitivity tests was conducted on key assumptions. These tests involved systematic variation of core model parameters within empirically plausible ranges:

- **Interest rate cap:** 1.6 % to 2.2 % (baseline: 2.0 %)
- **Loan approval sensitivity (β_1, β_2):** ±15 % around baseline estimates
- **Developer price elasticity (α):** 0.8 to 1.5
- **Policy trigger threshold (price growth condition):** ±2 % variation in cumulative growth trigger

Each perturbation was applied independently across 100 Monte Carlo iterations per scenario, enabling assessment of both first-order effects and interaction variance across agent behaviors.

Key findings include:

- **The directionality and structure of major results—**such as subsidy-induced price inflation and credit-access stratification—remained consistent under all tested variations.
- **Price-to-income ratios (PIR)** fluctuated within a narrow band of ±0.7 from the baseline range (15.1–15.4), while **loan approval rates** varied by less than 3 % points in either direction.
- More aggressive elasticity assumptions (e.g., $\alpha = 1.5$) accelerated speculative cycles but did not alter the core feedback dynamics linking demand surges to price increases.

These outcomes suggest that the ABM framework is robust under reasonable parameter uncertainty, and that simulation results should be interpreted as scenario-based foresight rather than deterministic forecasts. The model's strength lies in revealing directional risks, threshold effects, and system fragility arising from the interaction between affordability policy, credit structures, and behavioral feedback loops.

4.6. Empirical validation of model outcomes

To assess the external validity and real-world applicability of the model's projections, key simulation outputs were compared against empirical benchmarks derived from government data and national housing market reports. Table 8 and Fig. 4 summarize this comparative validation.

This table presents observed values (e.g., PIR, loan approval rates, price growth rates) alongside simulation results for the same indicators under baseline policy assumptions. Deviations were generally within ±5 %, indicating strong model alignment.

Table 8
Comparison between simulated outputs and observed market indicators (Taiwan, 2023).

Output indicator	Model result (range)	Observed benchmark	Source	Output indicator
Price-to-Income Ratio	15.1–15.4	15.2	Dept. of Land Admin, 2023	Price-to-income ratio
Loan approval rate	44 % – 51 %	45 % – 50 %	Central Bank, 2022–2023	Loan approval rate
Demand surge effect	+6.8 % price rise	+7 % (Q3 2023)	TWHUD, 2023	Demand surge effect

This figure contrasts three key indicators—price-to-income ratio, loan approval rate, and demand surge effect—between the ABM (yellow bars) and actual observed values (orange bars) from Taiwan's 2022–2023 housing data. The close match supports the behavioral accuracy of the model's agent rule and policy logic.

The tight alignments between simulated and real-world outcomes validate both the behavioral assumptions embedded in the agent's decision rules and the parameter calibration procedures discussed in Section 3.5. The model successfully reproduces macro-financial trends observed during the early rollout of the NYHLP, including:

- Increased approval rates following subsidy implementation
- Increased PIR levels in metropolitan areas
- Demand surge concentrated among youth and first-time buyers

This empirical cross-validation enhances confidence in the model's predictive capacity and strengthens the credibility of forward-looking simulations under future policy configurations. It also reinforces the model's value as a decision-making tool for evaluating potential side effects and trade-offs in evolving housing policy regimes.

5. Discussion and policy integration

This section critically examines the effectiveness of Taiwan's NYHLP by integrating simulated policy scenarios and agent interactions. Through dynamic modeling of homebuyer behavior, loan policies, government subsidies, and developers' responses, the analysis identifies both intended outcomes and unintended consequences of the program. Additionally, the discussion offers actionable policy recommendations to enhance housing affordability and market stabilization, while situating Taiwan's experience within the broader global housing policy discourse.

5.1. Policy effectiveness in context

Simulation results confirm that the NYHLP has successfully reduced entry barriers for young and lower-income buyers, demonstrating short-term gains in homeownership access. These findings are consistent with evidence from South Korea and the United States, where targeted subsidies and low-interest loans similarly expanded housing access (Kang & Park, 2024; Johnson & Smith, 2023). However, the model also reveals significant side effects: demand stimulation, when unaccompanied by timely supply-side expansion, leads to upward pressure on property prices—especially in supply-constrained urban areas.

This dynamic, observed in markets such as Singapore and Mainland China, reinforces the importance of policy coordination (Galster & Lee, 2020; Zhang & Li, 2019a, 2019b). The models' outputs illustrate a reinforcing feedback loop: subsidies lower borrowing costs → more buyers enter the market → developers raise prices → affordability gains

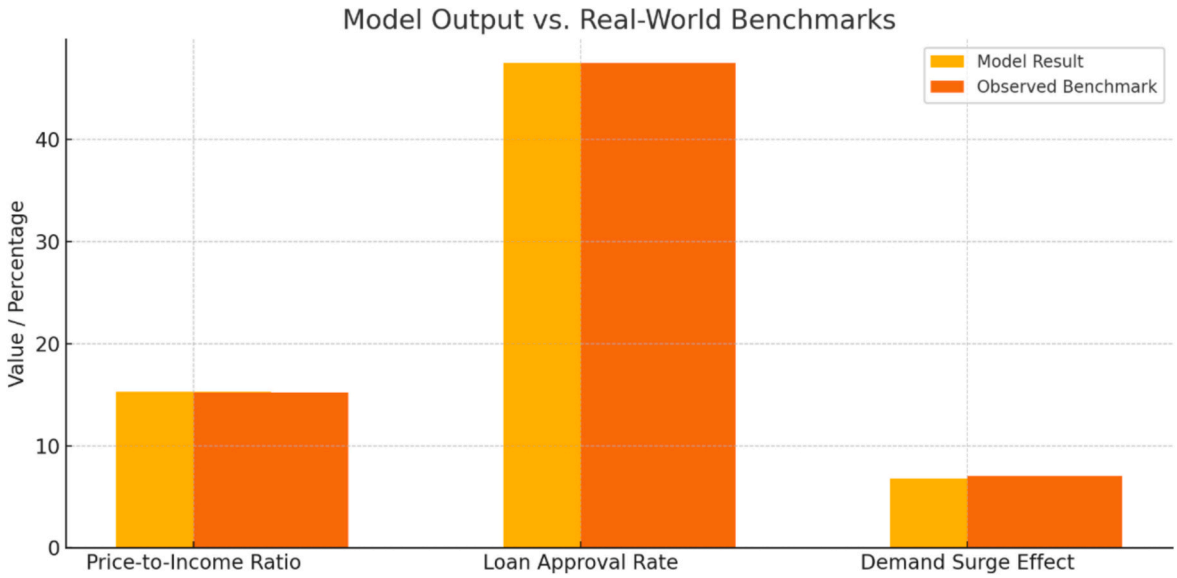

Fig. 4. Comparison of simulated outputs and observed real-world benchmarks across key housing indicators.

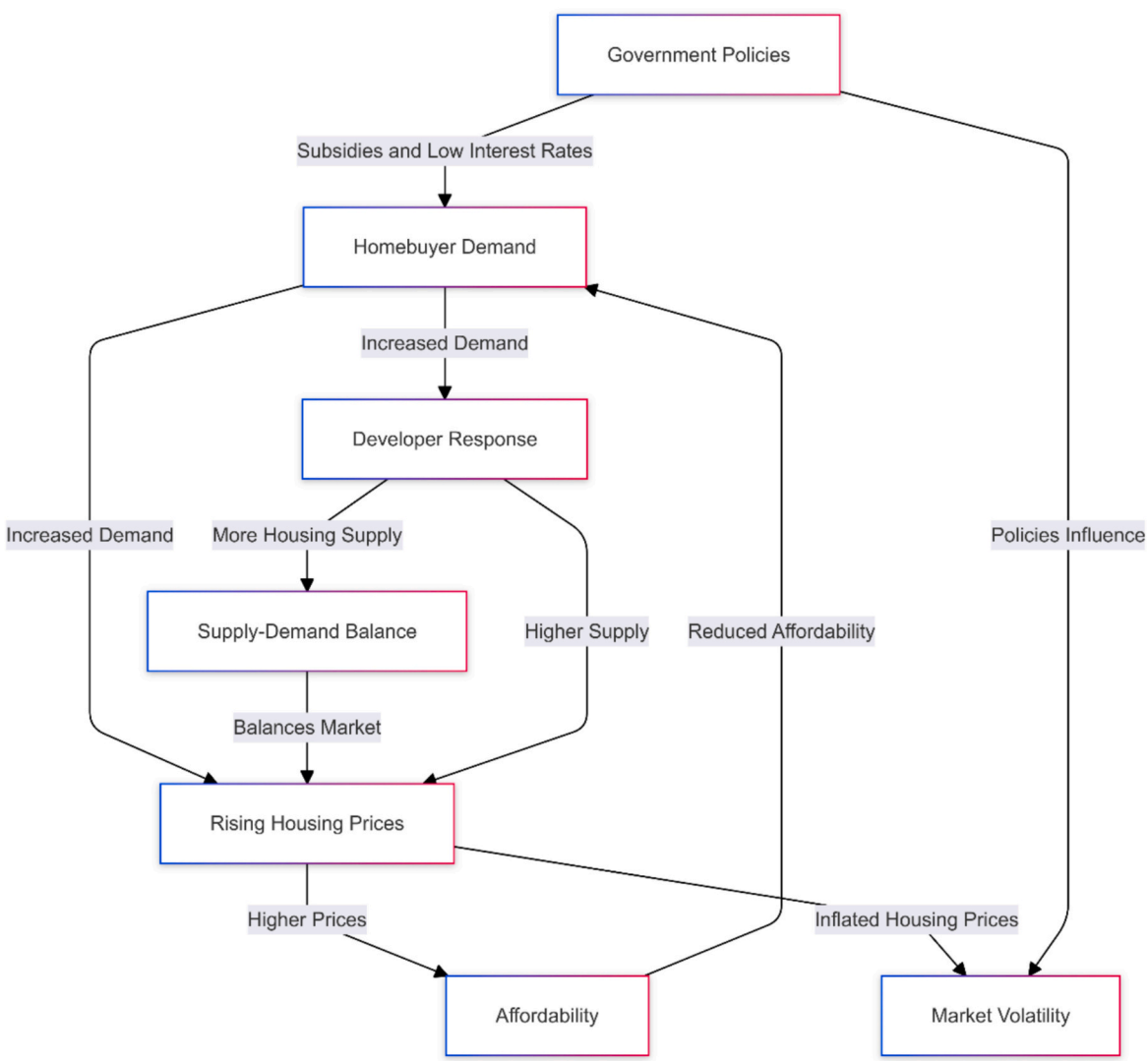

Fig. 5. Causal loop diagram of policy impact on housing market.

are diluted.

Importantly, the model functions not as a predictive tool but as a forward-looking framework for identifying risk patterns. It highlights early-warning signals—such as inflationary pressures and supply-demand imbalances—allowing for pre-emptive policy calibration before structural distortions materialize (Fig. 5).

This diagram visualizes the interplay between demand-side interventions and price feedback. It underscores how policies aimed at improving affordability can, if uncoordinated with supply expansion, create feedback-driven volatility and affordability traps.

5.2. Key driver of shaping homebuyer behavior

Simulation results point to three core factors that shape buyer behavior: price expectations, interest rates, and subsidy structure. Their interaction produces non-linear effects, particularly for lower-income homebuyer groups.

Price expectations are highly influential. When buyers anticipate further price increases—often prompted by time-bound subsidies—they are incentivized to enter the market sooner. This behavior, consistent with behavioral economic literature, can amplify short-term demand and accelerate inflationary cycles if not offset by new housing supply.

Interest rates remain a central lever. Lower rates increase loan approvals and accessibility for credit-constrained groups. In contrast, a modest increase can reduce approval rates sharply, disproportionately affecting financially vulnerable applicants. These findings position the interest rate as both an affordability tool and an equity concern.

Subsidy structure introduces distributional bias. As shown in Fig. 4, higher-income buyers benefit more from current subsidies due to higher approval rates and stronger financial profiles. In contrast, low-income groups experience limited gains unless gain unless target adjustments—such as down payment relief or grace periods—are implemented (Fig. 6).

This figure compares the equity outcomes of existing versus progressive subsidy designs. It demonstrates that under the current mechanism, benefits are skewed toward middle- and high-income groups. A progressive subsidy scheme would improve approval for low-income buyers without reducing overall program effectiveness.

5.3. Policy recommendations and theoretical implications

To ensure empirical grounding, the following recommendations are directly derived from scenario simulations (Sections 4.2–4.6). Each proposal is aligned with current policy trajectories in Taiwan and relevant theoretical frameworks.

1. Institutionalize adaptive policy triggers to respond to market volatility.

Static subsidy programs risk exacerbating affordability issues during inflationary cycles. In Scenario 2, a + 6.8 % price surge occurred despite declining approval rates, driven by latent demand and bank caution. Embedding adaptive triggers, such as automatic subsidy adjustments when price exceeds 8 % growth over two quarters, would enhance real-time responsiveness and reduce inflationary spillovers.

2. Align demand-side measures with enforceable supply-side commitments.

All simulation scenarios confirm that demand-side stimulation, when decoupled from housing supply, intensifies speculative inflation. Price-to-income ratios consistently exceeded 15, even with strong subsidies.

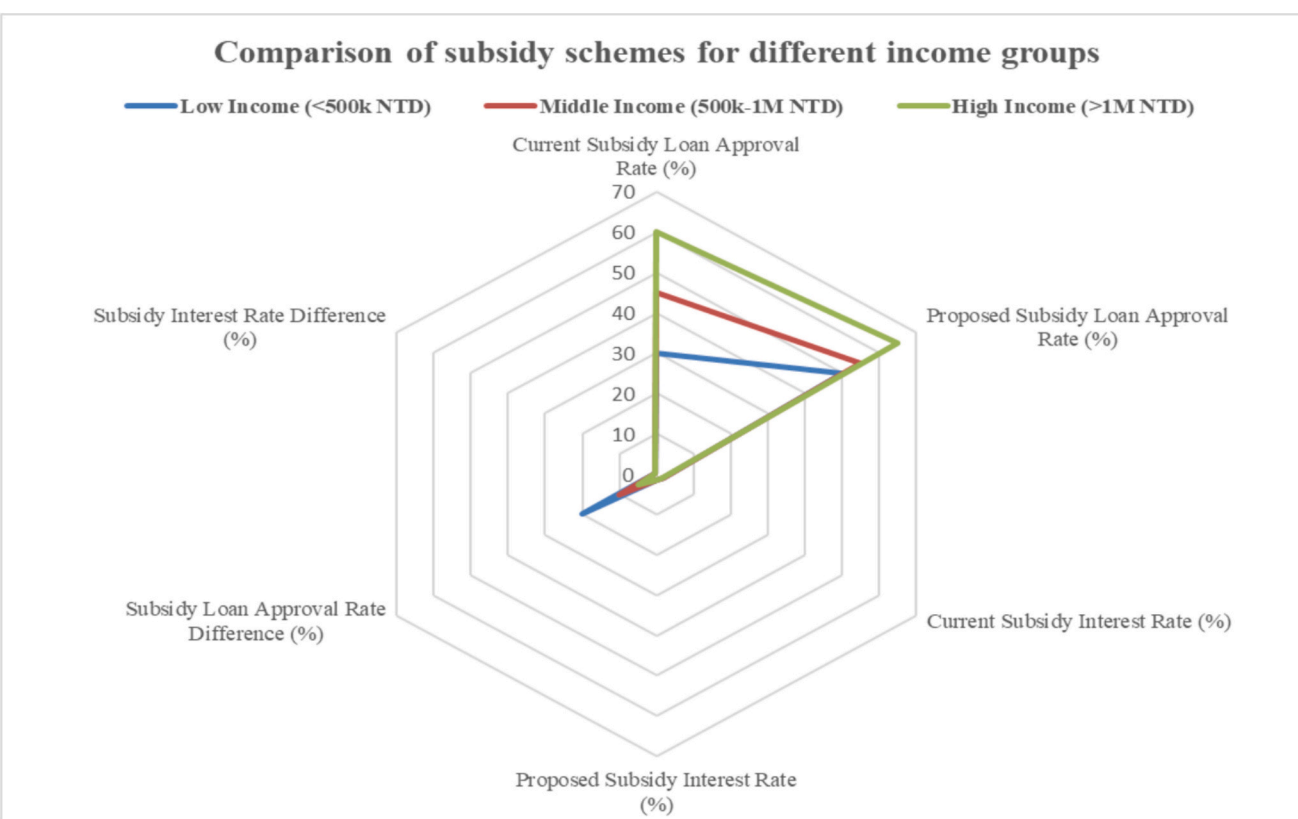

Fig. 6. Targeted subsidy impact heat map.

A more effective policy mix should include land-use reforms, incentives for green and mixed-income developments, and supply mandates for local governments to ensure synchronized interventions.

3. Redesign subsidy schemes to correct approval bias and promote equity.

As illustrated in Fig. 4, the simulation closely mirrors real-world trends, where middle-income applicants (income deciles M3–M5) are most likely to obtain loan approval, while low-income agents are often rejected, even under existing subsidy regimes. This pattern reflects a systemic mismatch between subsidy generosity and the eligibility criteria used by lending institutions. To correct this, policymakers should consider implementing progressive interest subsidies (e.g., inversely scaled by income), targeting down-payment assistance for credit-constrained households, and differentiated loan-to-value caps to discourage speculative multi-home purchases.

4. Increase transparency and consistency in loan approval mechanisms.

The model's logistic credit filter—calibrated using FSC data—revealed sharp discontinuities in approval rates across income bands. These findings echo real-world concerns regarding prolonged processing times, opaque rejections, and institutional discrimination. To address these issues, regulators should mandate public disclosure of approval metrics, conduct algorithmic bias audits, and establish centralized reporting systems to monitor credit access across demographic groups. Such reforms would help ensure that credit constraints do not disproportionately hinder housing access for youth and marginalized applicants.

5. Design programs with temporal safeguards to avoid policy cliffs.

Scenario simulations indicate that time-limited subsidies, such as those modeled in Scenario 3, can lead to accelerated purchasing followed by market dislocation if not paired with structured transition mechanisms. As of mid-2025, banks in Taiwan have reportedly begun adjusting internal NYHLP lending targets in anticipation of the program's expiration in July 2026 (China Times, July 23, 2025). Without clear exit strategies (e.g., step-down tapering or extension conditions), such "policy cliffs" may induce panic buying or abrupt market contractions. Programs should include phased withdrawal provisions calibrated to both fiscal cycles and prevailing market conditions.

These recommendations collectively reflect a central insight from this study: housing affordability is not a static outcome but a dynamically co-produced system, influenced by the timing, targeting, and interaction of fiscal, monetary, and regulatory policies. Well-intended interventions can produce adverse effects when implemented in isolation or misaligned with prevailing credit conditions, as evidenced by the model's simulation of feedback-driven affordability traps. This theoretical positioning directly challenges the assumptions embedded in DSGE frameworks such as Iacoviello and Neri (2010). Whereas DSGE models rely on rational expectations and static equilibrium adjustment, our ABM demonstrates that affordability outcomes are shaped by recursive, non-equilibrium processes. In particular, DSGE formulations treat subsidies or credit conditions as exogenous shocks, anticipating proportional spillovers that eventually converge to equilibrium. By contrast, our ABM embeds these interventions as endogenous, path-dependent triggers that interact with heterogeneous agents and institutional delays. This contrast reveals why DSGE models systematically underestimate the fragility of housing systems: they cannot capture how sequential interventions—credit loosening followed by regulatory tightening—generate oscillatory cycles of demand surges, price inflation, and mortgage withdrawal failures. The "feedback-driven affordability trap" thus reframes affordability not as a comparative statics problem but as an evolving dynamic system in which institutional misalignment can create reinforcing exclusion rather than convergence.

Recent official policy developments further corroborate this argument. On September 1, 2025, Taiwan authorities (Executive Yuan and Financial Supervisory Commission) announced that NYHLP mortgages would no longer be counted toward the statutory ceiling under Article 72–2 of the Banking Act, thereby expanding lending capacity for first-time, self-occupying buyers (Executive Yuan, 2025). While safeguards were introduced—including a one-loan-per-person limit, mandatory self-occupancy declarations, and stricter post-loan audits—the adjustment effectively "opened the credit valve" in ways that amplify the recursive risks highlighted by our model. Demand-side credit liberalization, when not matched with enforceable supply-side measures, risks entrenching the very affordability traps policymakers seek to resolve. DSGE equilibrium models, predicated on proportional and eventually stabilizing adjustments, are structurally unable to anticipate such path-dependent fragilities. By contrast, the ABM framework theorizes these recursive feedbacks, providing a more robust conceptual toolkit for understanding affordability crises under volatile policy regimes.

This systems-oriented view lays the foundation for the theoretical discussions in Section 6.2, which elaborates on how fragmented policy designs, under volatile macroeconomic conditions, can unintentionally reinforce exclusionary dynamics—even when aimed at improving affordability. This dynamic logic provides a conceptual bridge to the theoretical generalization developed in Section 6.2.

5.4. Comparative policy scenarios

The comparative simulation highlights how varying combinations of subsidies, interest rates, and credit regulations produce distinct market outcomes. As shown in Table 4, the "High Subsidy + Low Interest" scenario generates the strongest housing and price escalation, prompting developers to increase supply. Conversely, when subsidies are reduced and interest rates rise, demand weakens, prices decline, and supply contracts.

The "High Subsidy + Strict Lending" configuration represents a middle path: demand stabilizes, price growth flattens, and developer activity remains steady. This suggests that lending criteria can serve as a useful policy lever—containing inflationary pressures without fully dampening buyer participation.

These divergent outcomes indicate that no single policy variable determines affordability or market stability in isolation. Instead, effective interventions emerge from the coordinated interaction of fiscal support, credit regulation, and developer incentives. This simulation thus offers a framework for adaptive, scenario-based policy design that accommodates shifting economic and demographic conditions (Table 9).

6. Conclusion

This study underscores the critical role of government interventions, such as subsidies and interest rate reductions, in addressing housing affordability, particularly for low- and middle-income buyers. Taiwan's NYHLP demonstrates how targeted policies can alleviate financial burdens for first-time homebuyers while potentially serving as a model for

Table 9
Market predictions across different scenarios.

Scenario	Market demand	Housing price trend	Developer supply response
High subsidy, low interest rates	Strong demand, more buyers	5 %–10 % price increase	Increase in housing supply
Reduced subsidies, rising interest rates	Reduced demand, fewer buyers	5 %–7 % price decrease	Decrease in housing supply
High subsidy, strict lending	Moderate demand, stable buyers	Stable prices	Steady supply response

other developing regions. By balancing demand-side incentives with supply-side strategies, governments can enhance housing accessibility, foster sustainable economic growth, and promote social equity. Additionally, integrating environmental sustainability into housing policy can support long-term economic and ecological resilience.

6.1. Summary of findings

To consolidate the simulation results, this subsection synthesizes the key findings and assesses their implications for policy responsiveness and market stability.

This study provides a predictive assessment of Taiwan's NYHLP, using ABM to analyze the early-stage effects of housing policy interventions. Unlike retrospective studies based on long-term empirical data, this simulation-based approach offers forward-looking insights into how various combinations of subsidies, interest rate policies, and credit regulations may shape market behavior. By modeling the interactions among key stakeholders, including homebuyers, developers, banks, and government authorities, the study contributes to a more dynamic understanding of policy outcomes under evolving market conditions.

The results suggest that demand-side measures, such as interest rate subsidies and extended loan terms, can effectively improve housing access for young and lower-income buyers. These policies tend to increase loan approval rates and encourage market participation. However, a critical limitation emerges: rising demand in supply-constrained urban areas can prompt developers to raise prices, thereby reducing the long-term affordability gains such policies aim to provide. This highlights the importance of aligning financial incentives with supply-side responses.

Additionally, the simulation identifies interest rate policy as a primary determinant of both accessibility and pricing trends. Lower rates generally enhance inclusivity but may also generate inflationary pressure if not matched by increased housing supply. Conversely, higher rates reduce inflationary risk but tend to exclude financially vulnerable buyers. Taken together, these findings highlight the need for calibrated interventions that manage both demand stimulation and supply responsiveness.

It is important to emphasize that the simulation is not intended to deliver definitive forecasts. Rather, it serves as a scenario-based planning tool designed to help policymakers detect emerging risks, anticipate unintended consequences, and refine strategies in real time. This forward-looking perspective is especially valuable in housing markets characterized by volatility or limited data availability.

6.2. From simulation to theory: feedback-driven affordability traps and policy misalignment

Building on the simulation results, this section theorizes how affordability crises can emerge from fragmented or poorly timed interventions—a phenomenon referred to here as the feedback-driven affordability trap. By situating these dynamics within the broader literature on macro-policy coordination failure, the study highlights how interaction among fiscal, monetary, and regulatory instruments may inadvertently undermine housing stability. In doing so, it makes three key contributions to the fields of housing economics and urban policy simulation: advancing theoretical insight, offering empirically grounded interpretations, and demonstrating methodological innovations suited to dynamic and data-scarce policy environments.

1. Theoretical contribution

This study advances theoretical understanding of affordability dynamics by modeling how demand-side interventions, such as youth-targeted interest subsidies, interact endogenously with market-side developer responses and credit allocation behaviors. The model shows that subsidy-induced demand surges can lead to price inflation, particularly in a supply-constrained market, thereby reinforcing what this study terms a "feedback-driven affordability trap."

This concept extends earlier housing policy literature (e.g., Glaeser & Gyourko, 2003; Iacoviello & Neri, 2010) by embedding policy-induced shocks within a recursive, agent-based framework that reveals system-level vulnerability. In doing so, it bridges the gap between static equilibrium models—which often assume rational, price-clearing markets—and complex adaptive systems where institutional timing, credit flows, and household expectations interact dynamically.

A key theoretical innovation is the study of the introduction of a recursive feedback mechanism, wherein subsidy uptake increases loan approval, which boosts aggregate demand $D(t)$, ultimately influencing pricing $P(t)$ through developer behavior. This illustrates how policy objectives and market response interact non-linearly within a self-reinforcing loop. Such emergent patterns help conceptualize affordability traps—an outcome under-theorized in static or equilibrium housing frameworks. The study thus contributes a process-oriented, interactive theory of policy-induced market shifts and presents a replicable model structure for future ABMs studies of urban affordability.

Unlike models that treat government intervention as an exogenous parameter shift, this study conceptualizes housing policy as a temporally embedded, path-dependent process, one whose impacts evolve through agent responses and macro-financial conditions. This framing is especially pertinent to volatile housing markets, where incomplete coordination between regulatory authorities, credit institutions, and fiscal agencies often leads to suboptimal or contradictory outcomes.

This theoretical positioning directly challenges the assumptions embedded in DSGE frameworks such as Iacoviello and Neri (2010). Whereas DSGE models rely on rational expectations and static equilibrium adjustment, our ABM demonstrates that affordability outcomes are shaped by recursive, non-equilibrium processes. In particular, DSGE formulations treat subsidies or credit conditions as exogenous shocks, anticipating proportional spillovers that eventually converge to equilibrium. By contrast, our ABM embeds these interventions as endogenous, path-dependent triggers that interact with heterogeneous agents and institutional delays. This contrast reveals why DSGE models systematically underestimate the fragility of housing systems: they cannot capture how sequential interventions—credit loosening followed by regulatory tightening—generate oscillatory cycles of demand surges, price inflation, and mortgage withdrawal failures. The "feedback-driven affordability trap" thus reframes affordability not as a comparative statics problem but as an evolving dynamic system in which institutional misalignment can create reinforcing exclusion rather than convergence.

Recent policy adjustments further corroborate this argument. On September 1, 2025, Taiwan authorities announced that NYHLP mortgages would be exempt from the statutory ceiling under Article 72–2 of the Banking Act, thereby expanding lending capacity for first-time, self-occupying buyers (Executive Yuan, 2025). While safeguards were introduced—including a one-loan-per-person limit, mandatory self-occupancy declarations, and stricter post-loan audits—the adjustment effectively "opened the credit valve" in ways that amplify the recursive risks highlighted by our model. Demand-side credit liberalization, when not matched with enforceable supply-side measures, risks entrenching the very affordability traps policymakers seek to resolve. DSGE equilibrium models, predicated on proportional and eventually stabilizing adjustments, are structurally unable to anticipate such path-dependent fragilities. By contrast, the ABM framework theorizes these recursive feedbacks, providing a more robust conceptual toolkit for understanding affordability crises under volatile policy regimes.

This mechanism is not merely theoretical—it finds real-world validation in recent developments in Taiwan's property market. Crucially, these national-level policy adjustments resonate with localized market experiences. In 2023, a major residential project in southern Taiwan introduced a "daily pricing" strategy during its Phase I presale and quickly sold out amid optimistic sentiment and generous NYHLP

support. Encouraged by this success, the developer launched Phase II at higher prices. However, midway through the sales cycle, the central bank tightened credit policy, prompting banks to delay or halt mortgage approvals. This sudden reversal led to a wave of mortgage withdrawal failures, with early buyers losing deposits due to denied financing. The disconnect between subsidy-driven demand and contracting credit supply illustrated how fragmented institutional responses can entrench rather than relieve affordability constraints—validating the model's central premise regarding policy-induced fragility.

By integrating this empirical case into its simulation logic, the model broadens theoretical debates in housing economics beyond static affordability metrics. It aligns with the urban macroeconomic literature on coordination failures (e.g., Allen, McGregor, Swanstrom, & Wang, 2020; Iacoviello & Neri, 2010) and introduces a dynamic, feedback-sensitive framework that better explains why some affordability policies may unintentionally reinforce exclusion under unstable financial regimes.

2. Empirical contribution

This study presents a forward-looking scenario analysis of Taiwan's NYHLP, offering quantitative projections for policy configurations (2023–2026) during their early implementation phase. By testing dynamic variations in interest caps and subsidy amounts under differing market demand conditions, the model empirically demonstrates:

- The inflationary side effects of demand stimulation unaccompanied by supply-side expansion.
- The limited efficacy of credit policies in the absence of parallel supply governance.
- The nonlinear sensitivity of loan approval rates to evolving macroeconomic conditions.

Importantly, the model is calibrated using multi-source empirical data (e.g., central bank statistics, transaction-level benchmarks, and policy timelines), ensuring that outcomes reflect institutional realities and market dynamics. These findings provide timely empirical guidance to other urban housing systems—particularly in East Asia—that face similar challenges related to youth unaffordability and speculation pressure.

3. Methodological contribution

This study contributes to the ABM literature by integrating three mechanisms that are rarely combined in prior housing simulations:

Temporal policy triggers, where interventions (e.g., subsidy reductions) are activated based on real-time indicators such as cumulative price growth;

Agent-level financial thresholds, implemented via a logistic loan approval function that endogenously determined credit access based on income and interest rates;

Endogenous developer pricing, modeled through a demand–supply elasticity function, allows prices to respond flexibly to shifts in demand rather than follow preset schedules.

Compared to existing ABMs that treat policy as static or exogenous (e.g., Lee & Kim, 2020a, 2020b; Su & Qian, 2023), this study proposes a modular, adaptive architecture that improves alignment with real-time governance needs. The incorporation of Monte Carlo iterations and sensitivity diagnostics further enhances the model's explanatory scope and replicability across diverse national contexts.

6.3. Research limitations and future directions

This section integrates the simulation outcomes, theoretical contributions, and empirical observations into a cohesive framework. It also discusses broader implications for housing policy beyond Taiwan.

1. Simplified behavioral and macroeconomic assumptions

The simulation shows that aggressive credit stimulation—through low interest rates and generous subsidies—can temporarily improve loan access, especially for low- to middle-income youth households. However, when such demand-side incentives are not accompanied by supply-side expansion, they can rapidly trigger speculative price inflation, especially in inventory-constrained markets.

This demand-supply mismatch heightens systemic risks, particularly when developers respond to increased demand with opportunistic pricing strategies. This was modeled through the ABM's endogenous price adjustment function. At the same time, lenders tighten credit access in response to growing price volatility—mirroring behaviors observed during Taiwan's 2023–2024 NYHLP implementation.

2. Feedback-driven policy fragility: from simulation to theory

The model reveals that certain interventions may unintentionally create "feedback-driven affordability traps." These traps emerge when subsidies stimulate demand, triggering price escalations and subsequent credit tightening, which then reduce affordability—creating a cycle pattern of inclusion and exclusion.

This recursive mechanism supports and extends theories of coordination failure (e.g., Allen et al., 2020), credit procyclicality (Gyourko & Krimmel, 2021), and housing instability under asymmetric information (Glaeser et al., 2008). It challenges static housing models that assume frictionless institutional responses and underscores the importance of sequencing and responsiveness in public housing policy.

3. Empirical resonance: the Taiwan case

These dynamics are not abstract. In 2023, the initial rollout of the NYHLP in Kaohsiung triggered a presale boom. However, midway through the cycle, the central bank tightened credit policies, prompting banks to reject previously approved loans. As a result, many contracts were withdrawn and youth buyers lost deposits—providing real-world validation of the model's affordability trap logic. Recent reports (e.g., China Times, 2025-07-21) indicate that major banks have reduced NYHLP lending targets, citing institutional misalignment and tightening subsidy uncertainties.

Thus, the model serves not only as an academic framework but also as a practical diagnostic tool to identify risk before it becomes socially or politically entrenched.

6.4. Advancing simulation experiments

Looking beyond the NYHLP, this section underscores the broader utility of ABM in housing policy and suggests ways to improve its predictive value and relevance. This study demonstrates ABM's potential as a forward-looking tool to assess the impacts of youth housing policy. By simulating interactions among homebuyers, developers, lenders, and regulators, ABM enables the examination of emergent behaviors and unintended outcomes that traditional models may overlook. Its ability to model feedback loops and nonlinear effects makes it especially useful in markets marked by volatility, heterogeneity, and institutional complexity.

Nonetheless, the results should be seen as indicative rather than predictive. The model uses simplified assumptions about macroeconomic stability and agent behavior. It is best employed as a heuristic framework to help anticipate systematic risks, guide contingency planning, and inform adaptive policymaking—rather than as a forecasting instrument. The model highlights potential areas where policy interventions may lead to inflation, exclusion, or speculation.

Beyond its heuristic value, the ABM framework also departs from DSGE's equilibrium-centered paradigm by foregrounding feedback and heterogeneity. This distinction highlights ABM's capacity to reveal

evolving affordability traps that static equilibrium models cannot anticipate, reinforcing its methodological utility for housing policy analysis.

Future research should prioritize enhancing realism by integrating behavioral economics components (e.g., bounded rationality, shifting risk perception), accommodating regional shocks and demographic transitions, and validating results with newly available real-world data. Applying the model to other urban settings—especially those facing similar housing affordability pressure—would also help evaluate its generalizability and practical relevance.

Finally, this study reinforces the view that youth housing policies, when designed with adaptability and equity in mind, can serve not only as affordability instruments but also as catalysts for broader urban resilience. Linking simulation-based policy design with urban governance goals—such as social equity, sustainability, and demographic transition—can offer a more integrated strategy for long-term housing stability. This approach may inform policymakers not only in Taiwan but also in other cities seeking to implement responsive and context-sensitive housing solutions under uncertain macroeconomic trajectories.

CRediT authorship contribution statement

Kuo-Yan Wang: Writing – review & editing, Writing – original draft, Supervision, Methodology, Conceptualization. **Jing Yu:** Validation, Software, Resources, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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